

protege - Data Quality Report

Report date 2026-07-17 | delivery dates: 07-17-2026 | sessions checked: 3

Summary

1 check needs attention and 1 warning across 3 checked sessions. See the details section below for the affected sessions.

Quality checks at a glance

Check	Result	Sessions	Notes
Controls-to-video sync	Needs attention	0/3	Differs per session - see details below.
All recorded controls captured	Warning	0/3	Differs per session - see details below.
All required files delivered	Passed	delivery	Delivery: OK
Caption/annotation data delivered	Passed	3/3	All 3 sessions: per-frame action labels + key bindings -
Frame telemetry readable	Passed	delivery	Delivery: OK
Frame timestamps in order	Passed	3/3	Differs per session - see details below.
Key bindings readable	Passed	delivery	Delivery: OK
Processed without errors	Passed	3/3	All 3 sessions: OK
Session metadata follows the agreed format	Passed	delivery	Delivery: OK
Session metadata readable	Passed	delivery	Delivery: OK

Technical check ids (for engineers): Controls-to-video sync = action_video_grounding; All recorded controls captured = input_data_completeness; All required files delivered = required_game_artifacts_present; Caption/annotation data delivered = vendor_captions_present; Frame telemetry readable = frames_csv_schema_safe; Frame timestamps in order = timestamp_monotonicity; Key bindings readable = key_bindings_json_schema_safe; Processed without errors = normalization_error_free; Session metadata follows the agreed format = session_json_v31_contract; Session metadata readable = session_json_schema_safe

Session identifiers

Session	Your session id (as uploaded)	Game	Delivery
7fd1f3..72fbe2	2026-06-06T16-01-50Z_outerwild_c_82f637f618cc1ec0	Outer Wilds	07-17-2026
b6275e..6e17eb	2026-06-06T18-50-54Z_outer_wild_c_82f637f618cc1ec0	Outer Wilds	07-17-2026
e0cd79..87f90c	2026-06-06T13-59-38Z_kamla_c_82f637f618cc1ec0	Kamla	07-17-2026

Session = the abbreviated Odyssey session id used throughout this report; full ids appear in the raw lines under each finding.

Details

Controls-to-video sync - Needs attention

Whether recorded button presses line up with what happens on screen.

Technical id: action_video_grounding

- Session 7fd1f3..72fbe2 (Outer Wilds, 07-17-2026): video 151.5 ms behind inputs - outside the 150 ms tolerance (-4 frames at 26.397 fps over 3600 analyzed frames); measured from the mouse signal, active in 19% of frames; input-to-motion correlation -0.46 (negative: on-screen motion moves opposite to the recorded inputs - please confirm the mouse/stick axis convention used during capture) - Needs attention

raw: action_video_grounding | observed |lag|=152ms | threshold <= 150ms | session 7fd1f35c560bc61a084317852272fbe2 | your session id 2026-06-06T16-01-50Z_outerwild_c_82f637f618cc1ec0

- Session b6275e..6e17eb (Outer Wilds, 07-17-2026): video 207.7 ms behind inputs - outside the 150 ms tolerance (-5 frames at 24.068 fps over 3600 analyzed frames); measured from the mouse signal, active in 20% of frames; input-to-motion correlation -0.39 (negative: on-screen motion moves opposite to the recorded inputs - please confirm the mouse/stick axis convention used during capture) - Needs attention

raw: action_video_grounding | observed |lag|=208ms | threshold <= 150ms | session b6275ec0590c4f556b581d6c766e17eb | your session id 2026-06-06T18-50-54Z_outer_wild_c_82f637f618cc1ec0

- Session e0cd79..87f90c (Kamla, 07-17-2026): video 2326.9 ms behind inputs - outside the 150 ms tolerance (-61 frames at 26.216 fps over 3600 analyzed frames); measured from the mouse signal, active in 15% of frames; input-to-motion correlation -0.19 (negative: on-screen motion moves opposite to the recorded inputs - please confirm the mouse/stick axis convention used during capture) - Needs attention

raw: action_video_grounding | observed |lag|=2327ms | threshold <= 150ms | session e0cd798b8cf47f5dfa04e7c5ba87f90c | your session id

All recorded controls captured - Warning

Every keyboard, mouse, and gamepad input recorded in the session survives our processing, and each one is documented in the key-binding file.

Technical id: *input_data_completeness*

- Session 7fd1f3..72fbe2 (Outer Wilds, 07-17-2026): all 36 recorded controls captured; 5 undocumented (action 20, keyboard 15, mouse 1); not documented in the key-binding profile (checked against normalized_key_bindings): A, D, Enter, S, W - Warning

raw: *input_data_completeness* | observed all 36 recorded controls captured; 5 undocumented | threshold all recorded controls captured | session 7fd1f35c560bc61a084317852272fbe2 | your session id 2026-06-06T16-01-50Z_outerwild_c_82f637f618cc1ec0

- Session b6275e..6e17eb (Outer Wilds, 07-17-2026): all 31 recorded controls captured; 4 undocumented (action 17, keyboard 12, mouse 2); not documented in the key-binding profile (checked against normalized_key_bindings): A, D, S, W - Warning

raw: *input_data_completeness* | observed all 31 recorded controls captured; 4 undocumented | threshold all recorded controls captured | session b6275ec0590c4f556b581d6c766e17eb | your session id 2026-06-06T18-50-54Z_outer_wild_c_82f637f618cc1ec0

- Session e0cd79..87f90c (Kamla, 07-17-2026): all 15 recorded controls captured; 2 undocumented (action 7, keyboard 6, mouse 2); not documented in the key-binding profile (checked against normalized_key_bindings): Left, Right - Warning

raw: *input_data_completeness* | observed all 15 recorded controls captured; 2 undocumented | threshold all recorded controls captured | session e0cd798b8cf47f5dfa04e7c5ba87f90c | your session id 2026-06-06T13-59-38Z_kamla_c_82f637f618cc1ec0

Frame timestamps in order - Passed

Every video frame's timestamp moves forward in time, so the recording can be segmented reliably.

Technical id: *timestamp_monotonicity*

- Session 7fd1f3..72fbe2 (Outer Wilds, 07-17-2026): 33 ms median frame spacing; 1.24% of intervals outside 0.1x-3x the median (largest interval 366 ms) - under the 5% buffer - Passed

raw: *timestamp_monotonicity* | observed 33.0 | threshold >= 0 | session 7fd1f35c560bc61a084317852272fbe2 | your session id 2026-06-06T16-01-50Z_outerwild_c_82f637f618cc1ec0

- Session b6275e..6e17eb (Outer Wilds, 07-17-2026): 33 ms median frame spacing; 4.11% of intervals outside 0.1x-3x the median (largest interval 400 ms) - under the 5% buffer - Passed

raw: *timestamp_monotonicity* | observed 33.0 | threshold >= 0 | session b6275ec0590c4f556b581d6c766e17eb | your session id 2026-06-06T18-50-54Z_outer_wild_c_82f637f618cc1ec0

- Session e0cd79..87f90c (Kamla, 07-17-2026): 33 ms median frame spacing; 2.14% of intervals outside 0.1x-3x the median (largest interval 333 ms) - under the 5% buffer - Passed

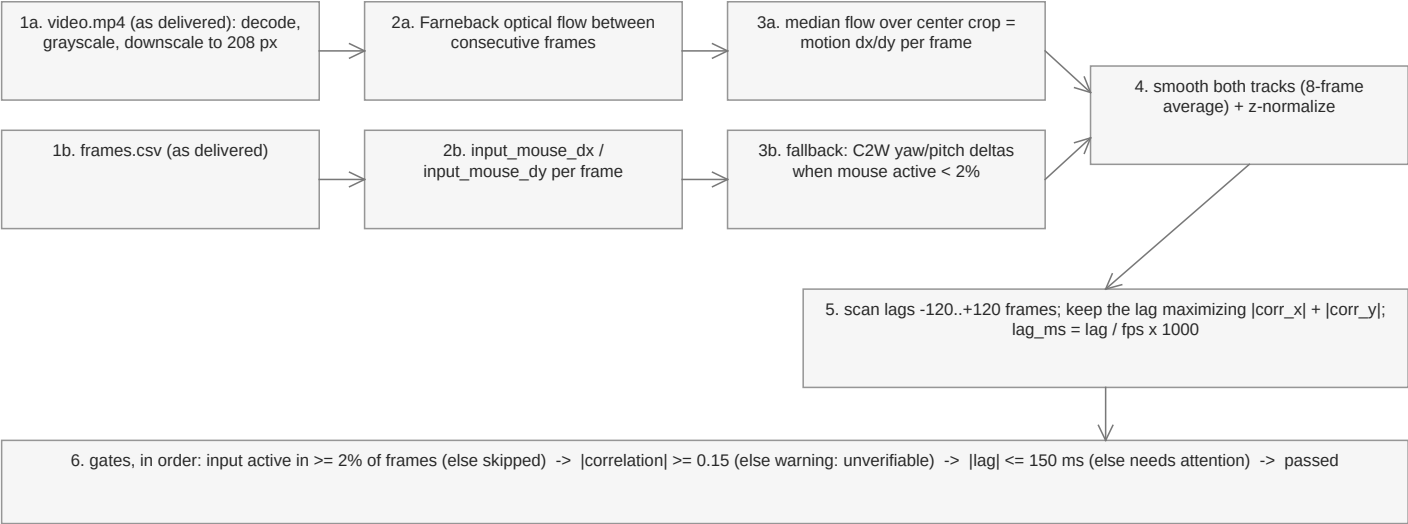
raw: *timestamp_monotonicity* | observed 33.0 | threshold >= 0 | session e0cd798b8cf47f5dfa04e7c5ba87f90c | your session id 2026-06-06T13-59-38Z_kamla_c_82f637f618cc1ec0

How we measure (for your engineers)

Each measurement below is computed from the files you delivered (video.mp4, frames.csv, key_bindings.json, session.json) - no Odyssey-internal data is involved, so you can replicate every reported number locally before or after upload.

Controls-to-video sync (action_video_grounding)

Reproducible from the delivered video.mp4 and frames.csv alone. (1) Video motion track: decode up to the first 3,600 frames, convert to grayscale, downscale to 208 px width; compute dense optical flow between consecutive frames (OpenCV calcOpticalFlowFarneback with pyr_scale 0.5, levels 3, winsize 15, iterations 3, poly_n 5, poly_sigma 1.2, flags 0); the per-frame motion value is the median flow x/y over the central crop (rows 1/6-5/6, cols 1/7-6/7), which excludes HUD overlays and the first-person viewmodel. (2) Input track: per-frame mouse deltas (input_mouse_dx / input_mouse_dy). When mouse deltas are active in under 2% of frames, camera yaw/pitch deltas derived from the C2W rotation columns are used instead; the report names which signal was used. (3) Both tracks are smoothed with an 8-frame moving average and z-normalized (zero mean, unit variance). For every candidate lag within +/-120 frames, the per-axis correlation is mean(input[t+lag] x motion[t]); the reported lag maximizes |corr_x| + |corr_y|, and lag_ms = lag_frames / fps x 1000 using the container frame rate (60 fps assumed when unreadable). Positive lag means the video runs ahead of the recorded inputs; negative means the video runs behind. The reported correlation is the stronger axis at the best lag, signed: a negative value means on-screen motion moves opposite to the recorded inputs (check the axis convention used during capture). Gates: input activity >= 2% of analyzed frames (below: check is skipped), |correlation| >= 0.15 to trust the estimate (below: warning, alignment unverifiable), |lag| <= 150 ms to pass. The recipe above is complete - it can be reimplemented with OpenCV and numpy alone. Reference code: this report runs compute_video_global_motion and estimate_action_video_lag in Odyssey's dq/checks/game/action_grounding.py; ask your Odyssey contact for the standalone script. Note that the shared vendor_qa.py pre-upload script covers file, schema, camera, and interval checks but not this sync measurement.



All recorded controls captured (input_data_completeness)

Every distinct input token observed in the delivered frames.csv (keyboard, mouse buttons, actions, gamepad) is compared against the tokens that survive our normalization and against the key-binding profile. Tokens missing after processing are reported as lost; tokens with no key-binding entry are reported as undocumented and listed by name. Undocumented tokens usually mean the key-binding file does not cover every control the capture recorded - including non-game inputs that leaked into the frame capture.

Frame timestamps in order (timestamp_monotonicity)

Computed from timestamp_ms in the delivered frames.csv. Timestamps must be non-decreasing; equal consecutive timestamps are tolerated. Interval regularity is scored against the session's median frame interval: an interval below 0.1x or above 3x the median counts as a gap/jitter event, and the fraction of such events is reported per session with a 5% warning buffer. This is the same definition and buffer as the timestamp_jitter check in the shared vendor_qa.py pre-upload script, so your pre-upload numbers and this report should match.

Acceptance tolerances

Input-to-frame temporal alignment	<= 50 ms target, <= 150 ms hard bound
Frame timestamps	Non-decreasing, one row per frame; <= 5% outside 0.1x-3x median
Keyboard/mouse clock skew	Zero (same clock source)
Session metadata	Complete per spec-v2 contract

A failing session is excluded and reported same-day; the rest of the delivery proceeds. Deliveries with a confirmed alignment failure are returned as a batch for recapture. Live view: Superset dashboard 'Raw Ingestion - Volumes & DQ'. Raw check results remain available to Odyssey Researchers in the data catalog.